Lesson 22: Fixed Point Theorems

Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

Contents

8 Fixed Point Theorems **2**

8.1 Banach Contraction Mapping Theorem 2

8.2 Brouwer Fixed Point Theorem 3

8.3 Schauder Fixed Point Theorem 4

8.4 Kakutani Fixed Point Theorem 4

9 Exercises **6**

10 Summary of Key Results **8**

8 Fixed Point Theorems

Fixed point theorems guarantee the existence (and sometimes uniqueness) of solutions to equations of the form $T(x) = x$. They are among the most powerful tools in analysis, with direct applications to existence and convergence theorems in RL.

8.1 Banach Contraction Mapping Theorem

Definition 8.1 (Contraction)**.** Let (X, d) be a metric space. A map $T\colon X \to X$ is a *contraction* (or *contraction mapping*) if there exists $\gamma \in [0, 1)$ such that

$$d(Tx, Ty) \leq \gamma\, d(x, y) \quad \text{for all } x, y \in X.$$

The constant γ is called the *contraction factor*.

Theorem 8.1 (Banach Contraction Mapping Theorem)**.** *Let (X, d) be a nonempty complete metric space and $T\colon X \to X$ a contraction with factor $\gamma \in [0, 1)$. Then:*

(i) T has a unique fixed point $x^* \in X$ (i.e., $Tx^* = x^*$).

(ii) For any $x_0 \in X$, the iterates $x_{n+1} = Tx_n$ satisfy $x_n \to x^*$.

(iii) The rate of convergence is geometric: $d(x_n, x^*) \leq \frac{\gamma^n}{1-\gamma} d(x_0, x_1)$.

Proof. **(i) and (ii): Existence and convergence.** Fix $x_0 \in X$ and define $x_{n+1} = Tx_n$. First,

$$d(x_{n+1}, x_n) = d(Tx_n, Tx_{n-1}) \leq \gamma\, d(x_n, x_{n-1}) \leq \cdots \leq \gamma^n d(x_1, x_0).$$

For $m > n$, by the triangle inequality:

$$d(x_m, x_n) \leq \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \leq \sum_{k=n}^{m-1} \gamma^k d(x_1, x_0) \leq \frac{\gamma^n}{1-\gamma} d(x_1, x_0).$$

Since $\gamma < 1$, $\gamma^n \to 0$, so (x_n) is Cauchy. By completeness, $x_n \to x^*$ for some $x^* \in X$.

Since T is continuous (in fact, Lipschitz), $Tx^* = T(\lim x_n) = \lim Tx_n = \lim x_{n+1} = x^*$.

Uniqueness. If $Tx^* = x^*$ and $Ty^* = y^*$, then $d(x^*, y^*) = d(Tx^*, Ty^*) \leq \gamma\, d(x^*, y^*)$, so $(1-\gamma) d(x^*, y^*) \leq 0$, giving $d(x^*, y^*) = 0$.

(iii): Rate of convergence. Letting $m \to \infty$ in the estimate above: $d(x^*, x_n) = \lim_{m\to\infty} d(x_m, x_n) \leq \frac{\gamma^n}{1-\gamma} d(x_1, x_0)$. □

Banach Contraction Mapping Theorem

In a complete metric space, a contraction has a unique fixed point, iterative computation converges from any starting point, and the convergence rate is geometric: $d(x_n, x^*) \leq \frac{\gamma^n}{1-\gamma} d(x_0, Tx_0)$.

Value Iteration Convergence

Let $\mathcal{S}$ be a finite state space and consider the Banach space $(V(\mathcal{S}), \|\cdot\|_\infty)$ of bounded value functions. The **Bellman optimality operator** $T\colon V(\mathcal{S}) \to V(\mathcal{S})$ is defined by

$$(TV)(s) = \max_{a \in \mathcal{A}} \Big[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V(s') \Big].$$

One verifies that T is a γ-contraction in $\|\cdot\|_\infty$:

$$\|TV - TW\|_\infty \leq \gamma \|V - W\|_\infty.$$

By the Banach Contraction Mapping Theorem:

1. There exists a unique optimal value function V^* with $TV^* = V^*$ (the Bellman equation).
2. **Value iteration** $V_{n+1} = TV_n$ converges to V^* from any initialization.
3. The convergence rate is $\|V_n - V^*\|_\infty \leq \frac{\gamma^n}{1-\gamma}\|V_1 - V_0\|_\infty$.

This is the *fundamental convergence guarantee* for value iteration. The same argument applies to the Bellman evaluation operator for policy evaluation, and to Q-value iteration.

8.2 Brouwer Fixed Point Theorem

Theorem 8.2 (Brouwer Fixed Point Theorem)**.** *Let $C \subset \mathbb{R}^n$ be a nonempty compact convex set and $f\colon C \to C$ a continuous map. Then f has a fixed point: there exists $x^* \in C$ with $f(x^*) = x^*$.*

Proof. We prove the theorem for $C = \overline{B}^n = \{x \in \mathbb{R}^n : \|x\| \leq 1\}$ (the closed unit ball). The general case follows since any nonempty compact convex set in $\mathbb{R}^n$ is homeomorphic to $\overline{B}^n$ (or a lower-dimensional ball, where the argument is analogous).

Proof by contradiction using the retraction argument. Suppose f has no fixed point, i.e., $f(x) \neq x$ for all $x \in \overline{B}^n$. Define $r\colon \overline{B}^n \to S^{n-1}$ (the boundary sphere) as follows: for each x, draw the ray from $f(x)$ through x; let $r(x)$ be the point where this ray hits S^{n-1}.

Explicitly, $r(x) = x + t(x)(x - f(x))$ where $t(x) \geq 0$ is chosen so that $\|r(x)\| = 1$. This leads to a quadratic equation in t which has a unique nonnegative solution depending continuously on x (since $f(x) \neq x$).

The map r is continuous, and for $x \in S^{n-1}$, we have $\|x\| = 1$ and the ray from $f(x)$ through x hits S^{n-1} at x itself (taking $t = 0$), so $r(x) = x$.

Thus $r\colon \overline{B}^n \to S^{n-1}$ is a *retraction*: a continuous map that is the identity on S^{n-1}.

Claim: No such retraction exists. This is the "no-retraction theorem," which can be proved via:

1. *Algebraic topology:* r would induce a surjection $r_*\colon \pi_{n-1}(\overline{B}^n) \to \pi_{n-1}(S^{n-1})$, but $\pi_{n-1}(\overline{B}^n) = 0$ (the ball is contractible) while $\pi_{n-1}(S^{n-1}) \cong \mathbb{Z}$ (for $n \geq 2$), contradiction.
2. *Smooth approximation + degree theory:* Approximate r by a smooth map; the degree of the restriction $r|_{S^{n-1}} = \mathrm{id}$ is 1, but the degree of any map extending to the ball must be 0.
3. *Combinatorial (Sperner's lemma):* Triangulate $\overline{B}^n$ and use Sperner's lemma to show that any continuous map satisfying the boundary condition must have a "completely labeled" simplex, from which a fixed point is obtained in the limit of refinements.

Each approach yields the contradiction. Hence the assumption that f has no fixed point is false. □

Remark 8.1. Brouwer's theorem is a pure existence result: it gives no algorithm for finding the fixed point (in fact, computing Brouwer fixed points is PPAD-complete). For algorithmic purposes, the Banach contraction theorem is far more useful when it applies.

8.3 Schauder Fixed Point Theorem

The Schauder theorem generalizes Brouwer's theorem to infinite-dimensional spaces. The key idea is to approximate a compact operator by finite-dimensional maps and apply Brouwer.

Definition 8.2 (Compact Operator)**.** A map $T\colon C \to X$ (where $C \subset X$) is *compact* if T is continuous and maps bounded sets to relatively compact sets (i.e., sets whose closure is compact).

Theorem 8.3 (Schauder Fixed Point Theorem)**.** *Let X be a Banach space, $C \subset X$ a nonempty closed bounded convex set, and $T\colon C \to C$ a compact map. Then T has a fixed point.*

Proof. Since $T(C)$ is relatively compact (as C is bounded and T is compact), $K = \overline{T(C)}$ is compact. For each $n \in \mathbb{N}$, cover K by finitely many balls of radius $1/n$: there exist $y_1^n, \dots, y_{k_n}^n \in T(C)$ with $K \subset \bigcup_{i=1}^{k_n} B(y_i^n, 1/n)$.

Define $C_n = \operatorname{conv}\{y_1^n, \dots, y_{k_n}^n\} \subset C$ (since C is convex and $y_i^n \in T(C) \subset C$). This is a compact convex subset of a finite-dimensional subspace.

Define $P_n\colon K \to C_n$ by

$$P_n(x) = \frac{\sum_{i=1}^{k_n} \lambda_i^n(x)\, y_i^n}{\sum_{i=1}^{k_n} \lambda_i^n(x)}, \quad \text{where } \lambda_i^n(x) = \max\bigl(0, 1/n - \|x - y_i^n\|\bigr).$$

This is the *Schauder projection.* Note that $\sum_i \lambda_i^n(x) > 0$ for $x \in K$ (since $K \subset \bigcup_i B(y_i^n, 1/n)$), so P_n is well-defined. Moreover, P_n is continuous, maps into C_n, and satisfies

$$\|P_n(x) - x\| \le 1/n \quad \text{for all } x \in K.$$

To see this: $\lambda_i^n(x) > 0$ only when $\|x - y_i^n\| < 1/n$, so $P_n(x)$ is a convex combination of points within distance $1/n$ of x, hence $\|P_n(x) - x\| < 1/n$.

Now consider $T_n = P_n \circ T|_{C_n}\colon C_n \to C_n$. This is a continuous self-map of a compact convex subset of a finite-dimensional space. By the Brouwer Fixed Point Theorem, there exists $x_n \in C_n$ with $T_n(x_n) = x_n$, i.e.,

$$x_n = P_n(Tx_n).$$

Then $\|x_n - Tx_n\| = \|P_n(Tx_n) - Tx_n\| < 1/n$.

Since (x_n) is a sequence in C and T is compact, (Tx_n) has a convergent subsequence. Along this subsequence, $x_n - Tx_n \to 0$ and $Tx_n \to y$, so $x_n \to y$ as well. Since C is closed, $y \in C$. By continuity of T: $Ty = \lim T(x_n) = y$. □

Remark 8.2. There are several variants of Schauder's theorem. The version above (sometimes called the Schauder–Tychonoff theorem for the compact convex set case) is the most commonly used. A more general version states: if C is a nonempty convex subset of a Banach space (not necessarily closed or bounded) and $T\colon C \to C$ is continuous with $T(C)$ relatively compact, then T has a fixed point. The proof is similar.

8.4 Kakutani Fixed Point Theorem

Definition 8.3 (Set-Valued Map / Correspondence)**.** A *set-valued map* (or *correspondence*) from a set X to a set Y is a map $\Phi\colon X \rightrightarrows Y$ that assigns to each $x \in X$ a subset $\Phi(x) \subset Y$. A *fixed point* of $\Phi\colon X \rightrightarrows X$ is a point x^* with $x^* \in \Phi(x^*)$.

Definition 8.4 (Upper Hemicontinuity)**.** A set-valued map $\Phi\colon X \rightrightarrows Y$ between topological spaces is *upper hemicontinuous* (u.h.c.) if for every open set $V \subset Y$, the set $\{x \in X : \Phi(x) \subset V\}$ is open in X. Equivalently (when Y is compact): if $x_n \to x$ and $y_n \in \Phi(x_n)$ with $y_n \to y$, then $y \in \Phi(x)$ (closed graph property when the image sets are compact).

Theorem 8.4 (Kakutani Fixed Point Theorem)**.** *Let $C \subset \mathbb{R}^n$ be a nonempty compact convex set and $\Phi\colon C \rightrightarrows C$ a set-valued map such that:*

(i) $\Phi(x)$ is nonempty and convex for every $x \in C$.

(ii) Φ is upper hemicontinuous.

Then Φ has a fixed point: there exists $x^* \in C$ with $x^* \in \Phi(x^*)$.

Proof. We prove this by approximation with single-valued maps and application of Brouwer's theorem.

Step 1: Approximate selections. For each $k \in \mathbb{N}$, we construct a continuous function $f_k\colon C \to C$ such that the graph of f_k is "close" to the graph of Φ.

For each k, cover C by finitely many open balls $B(x_i, 1/k)$, $i = 1, \dots, m_k$, with $x_i \in C$. Let $\{\lambda_i\}$ be a partition of unity subordinate to this cover. For each i, pick any $y_i \in \Phi(x_i)$. Define

$$f_k(x) = \sum_{i=1}^{m_k} \lambda_i(x)\, y_i.$$

This is continuous, and $f_k(x) \in C$ since $y_i \in C$ and C is convex.

Step 2: Apply Brouwer. By Brouwer's Fixed Point Theorem, each f_k has a fixed point $x_k \in C$ with $f_k(x_k) = x_k$.

Step 3: Pass to the limit. Since C is compact, (x_k) has a convergent subsequence $x_{k_j} \to x^*$. We claim $x^* \in \Phi(x^*)$.

Fix $\varepsilon > 0$. Since Φ is u.h.c. and $\Phi(x^*)$ is convex (hence, being a convex compact subset of $\mathbb{R}^n$, it equals the intersection of all open convex sets containing it), let V be an open convex set with $\Phi(x^*) \subset V$. By upper hemicontinuity, $\{x : \Phi(x) \subset V\}$ is open and contains x^*, so for large j, $\Phi(x_{k_j}) \subset V$ and $\|x_{k_j} - x^*\| < \varepsilon$.

For large $k_j > 1/\varepsilon$: $\lambda_i(x_{k_j}) > 0$ only when $\|x_{k_j} - x_i\| < 1/k_j < \varepsilon$, so $\|x_i - x^*\| < 2\varepsilon$. For such i, when j is large enough, x_i is close to x^*, so $\Phi(x_i) \subset V$ (by upper hemicontinuity), hence $y_i \in V$. Then $f_{k_j}(x_{k_j}) = x_{k_j}$ is a convex combination of y_i's in V. Since V is convex, $x_{k_j} \in V$.

Since V was an arbitrary open convex neighborhood of $\Phi(x^*)$, and $x_{k_j} \to x^*$, we get $x^* \in \overline{V}$ for every such V. Since $\Phi(x^*)$ is compact and convex, $x^* \in \bigcap\{\overline{V} : V \supset \Phi(x^*), V \text{ open convex}\} = \Phi(x^*)$. □

Nash Equilibrium Existence via Kakutani

Consider a finite n-player game. Player i has a finite action set A_i and a payoff function $u_i\colon \prod_j A_j \to \mathbb{R}$. A *mixed strategy* for player i is a probability distribution $\sigma_i \in \Delta(A_i)$ (the simplex over A_i).

The **best-response correspondence** of player i is

$$BR_i(\sigma_{-i}) = \arg\max_{\sigma_i \in \Delta(A_i)} \sum_{a \in A} \Big(\prod_j \sigma_j(a_j)\Big) u_i(a).$$

Define the joint best-response $\Phi(\sigma) = \prod_i BR_i(\sigma_{-i})$ on $C = \prod_i \Delta(A_i)$, which is a nonempty compact convex set in $\mathbb{R}^{\sum_i |A_i|}$.

One verifies:

1. Each $BR_i(\sigma_{-i})$ is nonempty and convex (it is the set of maximizers of a linear function over a simplex).

2. Φ is upper hemicontinuous (by Berge's Maximum Theorem: the payoff is continuous and the constraint set is constant).

By Kakutani's theorem, Φ has a fixed point σ^*, which is precisely a **Nash equilibrium**: every player is best-responding to the others. This is the original proof by Nash (1950).

Remark 8.3. The fixed point theorems form a hierarchy:

$$\text{Banach} \subset \text{Brouwer} \subset \text{Schauder} \quad \text{and} \quad \text{Brouwer} \subset \text{Kakutani}.$$

Banach gives uniqueness and algorithms (iterative computation); Brouwer/Schauder/Kakutani give only existence. The choice depends on the structure available:

Theorem	**Setting**	**Gives**
Banach	Complete metric, contraction	Unique FP + algorithm
Brouwer	Compact convex $\subset \mathbb{R}^n$, continuous	Existence
Schauder	Bounded closed convex, compact map	Existence
Kakutani	Compact convex $\subset \mathbb{R}^n$, u.h.c. correspondence	Existence

9 Exercises

Exercise 9.1 (Parallelogram Law Characterization)**.** Let $(X, \|\cdot\|)$ be a normed space satisfying the parallelogram law: $\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$ for all $x, y \in X$. Show that the formula

$$\langle x, y\rangle = \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2)$$

defines an inner product on X whose induced norm equals $\|\cdot\|$. *Hint:* Verify additivity $\langle x+z, y\rangle = \langle x, y\rangle + \langle z, y\rangle$ first for $z = x$, then for $z = y$, extend to rational scalars, then use continuity.

Exercise 9.2 (Non-Inner-Product Spaces)**.** Show that ℓ^1 and ℓ^∞ do not satisfy the parallelogram law, and therefore are not inner product spaces. Find explicit vectors x, y witnessing the failure.

Exercise 9.3 (Orthogonal Projection Computation)**.** Let $H = L^2([0,1])$ and $M = \operatorname{span}\{1, t, t^2\}$ (polynomials of degree ≤ 2). Compute the orthogonal projection of $f(t) = e^t$ onto M using:

(a) The Gram–Schmidt process applied to $\{1, t, t^2\}$.

(b) The normal equations $\langle f - Pf, g\rangle = 0$ for all $g \in M$.

Exercise 9.4 (Riesz Representation and Gradient)**.** Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be differentiable. The derivative $Df(x)$ at a point x is a bounded linear functional on $\mathbb{R}^n$. Use the Riesz Representation Theorem to show that there exists a unique $\nabla f(x) \in \mathbb{R}^n$ such that $Df(x)[h] = \langle \nabla f(x), h\rangle$ for all $h \in \mathbb{R}^n$. This justifies calling $\nabla f(x)$ "the gradient." How does the gradient depend on the choice of inner product?

Exercise 9.5 (Banach Contraction and Successive Approximation)**.** Consider the integral equation $x(t) = \int_0^t K(t,s)x(s)\,ds + g(t)$ on $C[0,1]$ with $\|K\|_\infty \leq M$. Show that the operator $T(x)(t) = \int_0^t K(t,s)x(s)\,ds + g(t)$ may not be a contraction in $\|\cdot\|_\infty$, but T^n is a contraction for large n, and deduce that T has a unique fixed point. *Hint:* Show $\|T^n x - T^n y\|_\infty \leq \frac{(Mt)^n}{n!}\|x-y\|_\infty$ and use the fact that a map with a contractive iterate has a unique fixed point.

Exercise 9.6 (Bellman Operator is a Contraction)**.** Let $\mathcal{S}$ and $\mathcal{A}$ be finite sets, $R\colon \mathcal{S}\times\mathcal{A} \to \mathbb{R}$ a reward function, $P(\cdot|s,a)$ a transition kernel, and $\gamma \in (0,1)$ a discount factor. Define the Bellman optimality operator T on $V(\mathcal{S})$ (bounded real-valued functions on $\mathcal{S}$ with the sup norm) by

$$(TV)(s) = \max_{a\in\mathcal{A}}\Big[R(s,a) + \gamma\sum_{s'} P(s'|s,a)V(s')\Big].$$

(a) Prove that T is a γ-contraction: $\|TV - TW\|_\infty \leq \gamma \|V - W\|_\infty$. *Hint:* For each s, let a^* achieve the max for $TV(s)$; then bound $TV(s) - TW(s) \leq \gamma \sum_{s'} P(s'|s, a^*)|V(s') - W(s')|$.

(b) Deduce from the Banach contraction theorem that value iteration converges geometrically and that V^* is the unique solution of the Bellman equation.

(c) How many iterations are needed to guarantee $\|V_n - V^*\|_\infty \leq \varepsilon$?

Exercise 9.7 (Closed Graph Theorem Application)**.** Let H be a Hilbert space and $T\colon H \to H$ a linear operator satisfying $\langle Tx, y\rangle = \langle x, Ty\rangle$ for all $x, y \in H$. Use the Closed Graph Theorem to show that T is bounded. *Hint:* Show that T has a closed graph by using the symmetry condition to identify the limit.

Exercise 9.8 (Uniform Boundedness Application)**.** Let (f_n) be a sequence in H^* (the dual of a Hilbert space H) such that $\sum_{n=1}^{\infty} |f_n(x)|^2 < \infty$ for every $x \in H$. Show that $\sup_n \|f_n\| < \infty$. *Hint:* The partial sums $S_N(x) = \sum_{n=1}^{N} |f_n(x)|^2$ are pointwise bounded; apply the Uniform Boundedness Principle to appropriate operators.

Exercise 9.9 (Kakutani and Two-Player Zero-Sum Games)**.** Let $A \in \mathbb{R}^{m\times n}$ be a payoff matrix for a two-player zero-sum game. Player 1 chooses $p \in \Delta_m$ (the m-simplex) and Player 2 chooses $q \in \Delta_n$, with payoff $p^\top Aq$ to Player 1.

(a) Define the best-response correspondences $BR_1(q)$ and $BR_2(p)$.

(b) Verify the hypotheses of Kakutani's theorem for the joint best-response $\Phi(p, q) = BR_1(q) \times BR_2(p)$ on $\Delta_m \times \Delta_n$.

(c) Conclude that a Nash equilibrium (saddle point) exists and relate this to the minimax theorem.

Exercise 9.10 (Schauder Theorem Application)**.** Let $K\colon [0, 1] \times [0, 1] \to \mathbb{R}$ be continuous and define $T\colon C[0, 1] \to C[0, 1]$ by $(Tf)(t) = \int_0^1 K(t, s)f(s)\, ds$.

(a) Show that T maps the closed ball $\overline{B}(0, R) \subset C[0, 1]$ into itself for $R \geq \|K\|_\infty \cdot R$ (choose R appropriately).

(b) Show that $T(\overline{B}(0, R))$ is equicontinuous (use uniform continuity of K).

(c) Apply the Arzelà–Ascoli theorem to conclude T is compact.

(d) Apply Schauder's theorem to find a fixed point.

10 Summary of Key Results

Result	**Statement (Informal)**	**RL/ML Connection**
Cauchy–Schwarz (Thm. ??)	$\lvert\langle x, y\rangle\rvert \leq \lVert x\rVert\lVert y\rVert$	Bounds on correlations; kernel evaluations
Parallelogram Law (Thm. ??)	$\lVert x+y\rVert^2+\lVert x-y\rVert^2 = 2(\lVert x\rVert^2+\lVert y\rVert^2)$	Characterizes inner product spaces
Projection Theorem (Thm. ??)	$H = M \oplus M^\perp$ for closed M	LSTD projection; least-squares approximation
Bessel's Inequality (Thm. ??)	$\sum \lvert\langle x, e_n\rangle\rvert^2 \leq \lVert x\rVert^2$	Energy bounds for Fourier coefficients
Parseval's Identity (Thm. ??)	$\lVert x\rVert^2 = \sum \lvert\langle x, e_n\rangle\rvert^2$ for ONB	Isometry between signal and coefficient spaces
Riesz Representation (Thm. ??)	Every $f \in H^*$ is $\langle \cdot, y\rangle$	RKHS theory; gradient = Riesz representative
Hahn–Banach (Thm. ??)	Linear functionals extend with norm preserved	Dual characterizations; separation theorems
Uniform Boundedness (Thm. ??)	Pointwise bounded $\Rightarrow$ uniformly bounded	Convergence of operator sequences
Open Mapping (Thm. ??)	Surjective bounded linear maps are open	Bounded inverse theorem
Closed Graph (Thm. ??)	Closed graph $\Leftrightarrow$ bounded (Banach spaces)	Proving operator continuity
Banach Contraction (Thm. 8.1)	Contraction has unique FP; iterates converge at rate γ^n	Value iteration; policy evaluation; Q-learning
Brouwer FPT (Thm. 8.2)	Continuous self-map of compact convex set has FP	Existence results in finite dimensions
Schauder FPT (Thm. 8.3)	Compact self-map of closed bounded convex set has FP	Existence of solutions to integral equations
Kakutani FPT (Thm. 8.4)	u.h.c. convex-valued correspondence has FP	Nash equilibrium existence

Key Takeaways:

1. **Inner products provide geometry.** The Cauchy–Schwarz inequality, projections, and orthogonal decompositions are all consequences of the inner product structure. This geometry is what makes Hilbert spaces the natural setting for least-squares and projection methods.

2. **Hilbert spaces are self-dual.** The Riesz Representation Theorem identifies H with H^*, vastly simplifying the theory compared to general Banach spaces.

3. **The four fundamental theorems are structural results** about Banach spaces. They guarantee that linear operators and functionals behave well: they can be extended (Hahn–Banach), pointwise bounds imply uniform bounds (UBP), surjections are open (OMT), and closed-graph operators are bounded (CGT).

4. **Fixed point theorems bridge analysis and algorithms.** The Banach contraction theorem provides both existence, uniqueness, and an algorithm. Brouwer/Schauder/Kakutani provide existence in settings where contractions are not available, notably for Nash equilibria.

5. **The hierarchy of fixed point theorems** matches the hierarchy of RL/ML problems: contraction for value iteration, Kakutani for Nash equilibria, Schauder for existence of solutions to functional equations.